

Maxwell Patterson

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SUMMARY

Senior Data Scientist who builds machine learning systems end to end- from predictive models and LLM-powered applications to the deployment and monitoring infrastructure that keeps them in production. Track record of converting models into business outcomes, including a safety model followed by the best incident-rate quarter in company history and analysis that surfaced ~\$25M in staffing overruns. People manager who developed a co-op into a full-time direct report, with experience communicating with stakeholders from field teams to executives.

TECHNICAL SKILLS

- **Machine Learning & AI:** Classification & regression (XGBoost, scikit-learn, tidymodels), deep learning (PyTorch), NLP, LLM application development, retrieval-augmented generation (RAG), agentic workflows & AI agent tooling (custom skills, adherence testing)
- **MLOps & Deployment:** Databricks (PySpark, Databricks SQL, Volumes, system tables), vetiver, Posit Connect, model monitoring, champion/challenger evaluation, Flask, FastAPI, Git
- **Programming:** Python (pandas, NumPy, scikit-learn, PyTorch, PySpark), R (tidyverse, tidymodels, Shiny), SQL (Databricks, PostgreSQL, SQL Server), JavaScript (Plotly, D3, Leaflet)
- **Analytics & Visualization:** Tableau, Looker, Shiny, dbt, Alteryx

PROFESSIONAL EXPERIENCE

SUFFOLK CONSTRUCTION

Senior Data Scientist

Data Scientist

Senior Data Analyst

Roxbury, MA

June 2025 – Present

August 2024 – June 2025

March 2022 – August 2024

- Trained and productionized a classification model identifying projects at risk of safety incidents, improving accuracy 17% over the prior version; the first full quarter after rollout was the best for incident rates in company history ([featured in a published Posit customer story](#))
- Built a stakeholder-facing model application surfacing predictions, lift charts, and performance monitoring, making model behavior legible to non-technical safety and operations leaders
- Established standardized ML deployment architecture for the team: versioned models with vetiver stored in Databricks Volumes, served via Posit Connect, with a champion/challenger framework for evaluating model updates
- Built a custom skills repository with automated tests for Devin, the company's agentic coding platform, including skills for spec-driven development and standardized Databricks connectivity; skill usage raised agent adherence to engineering standards from 60% to ~95%
- Designed and built an LLM-powered application that generates construction project schedules, combining agentic task generation with ML models for task duration estimation; now piloting with project teams
- Led analysis of project close-out staffing, revealing that close-out phases consistently consumed a higher share of staffing time than projected at project start, representing ~\$25M in staffing overruns and informing changes to workforce planning processes
- Mentored and managed a data science co-op through a summer internship and a graduate capstone project built on company data, then transitioned to serving as their direct manager upon their full-time hire

ZIPCAR

Data Analyst

Boston, MA

September 2021 – March 2022

- Diagnosed defects in a dbt-based lifetime value model, recovering ~\$70 in missing revenue per user for an affected member segment and improving reliability of revenue reporting
- Analyzed search behavior in key markets to identify disengaged members, producing retention insights adopted by product and marketing teams
- Built a Looker dashboard tracking billing accuracy and success KPIs for the billing team

NIELSENIQ

Analytical Framework Consultant

Chicago, IL

August 2019 – September 2021

- Led development of a Python/SQL reporting tool served from a FastAPI endpoint with built-in data validation, reducing manual analysis time by 95%
- Prototyped product databases for CPG retailers and manufacturers, working directly with client SMEs to confirm contractual requirements were met
- Served on an internal efficiency team replacing manual processes with Python automation to shorten product delivery timelines

REVENUE SOLUTIONS, INC.

IT Consultant / Associate IT Consultant

Chicago, IL

September 2016 – August 2019

- Provided client-site production support for SQL Server-based tax processing applications for state, city, and county agencies; recognized with a company award for client communication and relationship management
- Authored software requirements specifications for custom ETL interfaces across multiple tax types, enabling clients to integrate external data sources and improve tax processing accuracy

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Master of Science, Data Science

May 2024

- Coursework: Machine Learning, Deep Learning, NLP, Reinforcement Learning, Causal Inference, Advanced Predictive Models

NORTHWESTERN UNIVERSITY

Professional Certificate, Data Science & Data Visualization

February 2020

DICKINSON COLLEGE

Bachelor of Science, Mathematics and Physics

May 2016